Bekenstein-Hawking Entropy from Modular-Tensor-Category State Counting: Kaul-Majumdar $\mathrm{SU}(2)_k$ Closed Form, Tracked-Hypothesis Bundle for the General MTC, and Walker-Wang Anomaly Match for Anderson-Higgs Tetrad Substrates

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[1] *SK-EFT Hawking Project*

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We formalize the Bekenstein-Hawking entropy $S_{\mathrm{BH}} = A_H/(4G_N) - (3/2)\log[A_H/(4G_N)] + c_0$ at a black-hole horizon labeled by simple objects of a Modular Tensor Category (MTC), bridging the Anderson-Higgs tetrad-condensation (ADW) emergent-gravity program of Phase 6a with the Kaul-Majumdar $\mathrm{SU}(2)_k$ Verlinde-formula derivation [arXiv:gr-qc/0002040]. Following the Wave 3 deep-research literature survey, we ship in *Outcome-3 (tracked-hypothesis) mode* for the general-MTC case, with an *Outcome-2 (Conditional) sub-corollary* specialized to $\mathrm{SU}(2)_k$ under the Kaul-Majumdar derivation. Three load-bearing structural results: (i) the $-3/2$ log coefficient decomposes as $(-1/2)$ Gaussian-saddle + (-1) SU(2)-singlet-projection from the $I_0 - I_1$ cancellation; (ii) the leading $1/4$ prefactor is structurally a TUNING (Immirzi parameter γ), not a derivation, with two distinct counting prescriptions (Domagala-Lewandowski $\gamma_{\mathrm{DL}} \approx 0.2375$, Meissner $\gamma_M \approx 0.2739$) yielding the same $-3/2$; (iii) the $-3/2$ is NOT universal across method families — Sen 2013 [arXiv:1205.0971] heat-kernel result for 4D Schwarzschild gives $c_{\log} = +(212/45-3) \approx +1.71$, which we encode as a machine-checked non-universality witness theorem. We bundle the general-MTC tracked hypothesis as `H_HorizonBoundaryCondition` with five falsifier theorems (positivity, area-law leading scaling, second-law monotonicity, modular invariance, Walker-Wang anomaly match) and provide falsifier-instance status reports for the Fibonacci, Ising, $D(S_3)$, and toric-code MTCs (the abelian toric code immediately fails F2 since $\log d_{\max} = 0$). The synthesis "$\mathrm{SU}(2)_k$ MTC at the horizon of a 4D ADW substrate with Walker-Wang anomaly inflow from $\mathbb{Z}_2$ time-reversal symmetry" is unpublished as of April 2026 and is novelty-flagged accordingly.

I. INTRODUCTION

Phase 6a of the SK-EFT-Hawking program proves that linearized Einstein equations emerge from Anderson-Higgs tetrad condensation (Wave 1, `LinearizedEFE.lean`) with $G_N^{\mathrm{emerg}} = \alpha_{\mathrm{ADW}} \cdot 12\pi/(N_f\Lambda^2)$ [16, 17]. Bekenstein-Hawking entropy $S_{\mathrm{BH}} = A_H/(4G_N)$ is a constraint on the microscopic theory: any candidate emergent-gravity program must reproduce this thermodynamic law at the horizon.

The Kaul-Majumdar $\mathrm{SU}(2)_k$ Chern-Simons / Verlinde derivation [gr-qc/0002040] is the only equation-level closed form in the published literature yielding both the $1/4$ leading coefficient (after γ tuning) and the $-3/2$ logarithmic correction. Per the Wave 3 deep-research literature survey (`Lit-Search/Phase-6a/6a-Horizon MTC boundary conditions, Wave 3.md`, 2026-04-25), *no published paper places any specific finite MTC at the horizon of a 4D non-extremal black hole in an ADW or tetrad-condensate substrate*. The closest published MTC $\leftrightarrow$ BH-entropy identification is McGough-Verlinde for BTZ via the Virasoro modular S-matrix [arXiv:1308.2342]. We therefore ship `BHEntropyMicroscopic.lean` in tracked-hypothesis mode with a machine-checked Kaul-Majumdar $\mathrm{SU}(2)_k$ sub-corollary as the structural anchor.

Three structural results.

We establish three machine-checked structural facts:

(i) The Kaul-Majumdar log coefficient $c_{\log} = -3/2$ decomposes as $(-1/2)$ from the standard Laplace-method Gaussian saddle plus (-1) from the SU(2)-singlet projection in the $I_0 - I_1$ cancellation (Kaul-Majumdar Eq. (15)). The decomposition is a Lean theorem, `kaul_majumdar_log_decomposition`.

(ii) The leading $1/4$ prefactor. *In the LQG/Verlinde-counting route* the Immirzi parameter γ is fixed by demanding the counted leading area coefficient equal $1/(4G_N)$; distinct counting prescriptions yield distinct γ values (Domagala-Lewandowski $\gamma_{\mathrm{DL}} \approx 0.2375$ [arXiv:gr-qc/0407051]; Meissner $\gamma_M \approx 0.2739$ [arXiv:gr-qc/0407052]) with the *same* $-3/2$ log coefficient, and γ enters as an explicit field of the Lean structure `HorizonMTCBC` (§VI). The $1/4$ is, however, *not* irreducibly a tuning: in the ADW Sakharov induced-gravity substrate it is derived γ-independently (§IX) from the shared Seeley–DeWitt a_2 coefficient — the same a_2 that induces G_N also fixes the horizon entanglement-entropy area density, with the heat-kernel-to-Einstein–Hilbert ratio structurally equal to 4. The Immirzi value is then a *consistency check*, not the source of the $1/4$.

(iii) The $-3/2$ is NOT universal. Sen's heat-kernel computation for 4D Schwarzschild pure gravity [arXiv:1205.0971] gives $c_{\log} = (212/45 - 3) \approx +1.71$, an explicit disagreement with the Cardy-saddle anchor. We encode this as a non-universality witness theorem `sen_4d_disagrees_with_kaul_majumdar`, plus the sign

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theorem `sen_4d_log_coeff_pos`.

II. SETUP: MODULAR-TENSOR-CATEGORY HORIZON DATA

A. The HorizonMTCBC structure

A Modular Tensor Category at the horizon is encoded as a Lean structure `HorizonMTCBC` with fields: finite simple-object label set $\{0, 1, \ldots, N-1\}$ (with $0 =$ unit); per-object positive quantum dimension d_a; an attainable maximum $d_{\max}$; per-object area contribution $A_{H\,a} = 8\pi\gamma\ell_P^2\sqrt{C_2(a)}$ (in the $\mathrm{SU}(2)_k$ specialization); a dominant simple object; and the Immirzi tuning parameter $\gamma > 0$. The MTC categorical data (modular S, T matrices; F, R braided structure) is supplied per-instance by the Lean MTC stack: `SU2kFusion`, `FibonacciMTC`, `IsingBraiding`, `S3CenterAnyons`, `SphericalCategory`.

The total quantum dimension squared $\mathcal{D}^2 = \sum_a d_a^2$ is positive since the unit object contributes $1^2 = 1$ (`globalDimSq_pos`). The $\log d_{\max} \geq 0$ structural anchor follows from $d_{\max} \geq d(\text{unit}) = 1$ (`log_d_max_nonneg`). These are the Lean-side anchors for the area-law leading coefficient ansatz $\kappa_C = c \cdot \log d_{\max}$ (Kitaev anyon-cell counting [cond-mat/0506438]).

B. The Verlinde formula at the horizon

A horizon S^2 with p punctures carrying simple-object labels $a_1, \ldots, a_p$ has Verlinde-counted Hilbert-space dimension

$$\dim \mathcal{H}_{S^2;a_1,\ldots,a_p} \;=\; \sum_c \frac{S_{a_1c}\cdots S_{a_pc}}{(S_{0c})^{p-2}}, \tag{1}$$

where S is the modular S-matrix of the MTC. Equation (1) is the literal physical content of the horizon state count, encoded as `verlinde_dim_horizon` in `src.core.formulas`. We verify numerically that for $\mathrm{SU}(2)_k$ at $k = 2$ (Ising) the four-σ correlator gives $\dim = 2$ (the two Ising fusion channels), and the two-σ correlator gives $\dim = 1$ (charge conservation, σ self-dual).

III. THE KAUL-MAJUMDAR CLOSED FORM

A. The closed form

Kaul and Majumdar [1, 2] derived the $\mathrm{SU}(2)_k$ Verlinde-counted horizon entropy via the magnetic-quantum-number representation

$$\mathcal{N}_P = \sum_{m_l=-j_l}^{j_l} \left[\bar{\delta}_{\sum m,0} - \tfrac{1}{2}\bar{\delta}_{\sum m,1} - \tfrac{1}{2}\bar{\delta}_{\sum m,-1}\right] \tag{2}$$

followed by Laplace-method evaluation around $x = 0$ with $-F''(0) = \alpha\, A_H/4$. The $I_0 - I_1$ cancellation produces an extra inverse-Hessian factor, yielding

$$\boxed{S_{\mathrm{BH}} \;=\; \frac{A_H}{4\ell_P^2} - \frac{3}{2}\log\!\left(\frac{A_H}{4\ell_P^2}\right) + \text{const} + \mathcal{O}(A_H^{-1}).} \tag{3}$$

Equation (3) is encoded as `kaulMajumdarS`, with the structural log-coefficient extraction theorem

$$\texttt{kaulMajumdarS}(A,\, G,\, c_0) - A/(4G) - c_0 \;=\; -\tfrac{3}{2}\,\log[A/(4G)], \tag{4}$$

proven by `ring`.

B. The $-3/2$ decomposition

The structural decomposition $-3/2 = (-1/2) + (-1)$ has two parts: the arithmetic identity itself (encoded in Lean as `kaul_majumdar_log_decomposition`, which proves the rational identity $(-1/2) + (-1) = -3/2$ by `ring`) and the physical sourcing of each summand (documented at the prose level here, not yet formalized at the theorem level — i.e., there is no Lean theorem attaching the value $-1/2$ to a Gaussian-saddle object or -1 to a singlet-projection object):

$$c_{\log} \;=\; c_{\text{Gaussian}} + c_{\text{singlet}} \;=\; -\tfrac{1}{2} + (-1) \;=\; -\tfrac{3}{2}, \tag{5}$$

where $c_{\text{Gaussian}} = -1/2$ is the Laplace-method asymptotic $I_0 \sim C\, e^{F(0)}/\sqrt{-F''(0)}$ (the bounded-remainder content is captured by the `gaussianSaddleAsymptotic` theorem after Wave 6a.7 axiom retirement; cf. §III C) and $c_{\text{singlet}} = -1$ is the SU(2)-singlet projection from the $I_0 - I_1$ cancellation [1, 2]. Lifting the prose-level sourcing to per-summand Lean definitions (`gaussianContribution`, `singletProjectionContribution`) is a follow-on goal.

C. The Laplace-method bound (axiom retired Wave 6a.7; faithful definition + genuine $O(1/A)$ rate Wave 7C)

Wave 3 originally axiomatized the key bound:

$$\exists\, C > 0 : \;\Big|\texttt{verlindeEntropy\_SU2k}(A, G_N) - \texttt{kaulMajumdarS}(A, G_N \tag{6}$$

for all $A \geq 1$, $G_N > 0$. **Wave 6a.7 (2026-04-27) retired the axiom**; **Wave 7C (2026-06-14) made the definition *faithful***: $\texttt{verlindeEntropy\_SU2k}(A, G_N) :=$ $\texttt{continuumLogCatalan}(A/(8G_N \log 2))$ is now the *literal* log of the Γ-interpolated SU(2) singlet dimension $\Gamma(2x{+}1)/(\Gamma(x{+}1)\,\Gamma(x{+}2))$ — a genuine independent analytic object, *not* the earlier saddle self-definition $:= \texttt{kaulMajumdarS}(A, G_N, 0)$ (substrate-audit finding #13, "defining-the-conclusion"). The Kaul–Majumdar closed form is now a *derived corollary*: `gaussianSaddleAsymptotic` is the genuine

$O(1/A)$ rate $|\texttt{verlindeEntropy\_SU2k}(A, G_N) - \texttt{kaulMajumdarS}(A, G_N, \kappa_{\mathrm{KM}})| = O(1/A)$ with the literal nonzero constant $\kappa_{\mathrm{KM}} = \frac{3}{2}\log(2\log 2) - \frac{1}{2}\log\pi$ — no longer the vacuous $|x - x| = 0$. The literal $\mathrm{SU}(2)_k$ Verlinde-sum derivation of the $-3/2$ is the foundation: the horizon singlet count is the SU(2) singlet multiplicity = Catalan number $\binom{2m}{m}/(m+1)$ (the discrete $I_0 - I_1$), whose $-3/2$ follows from Mathlib's Stirling formula — NOT a Hardy–Ramanujan partition asymptotic (that is for *unrestricted* partitions; the horizon count is the *constrained* singlet count). The discrete asymptotic is sharpened to the $O(1/m)$ *rate* (`LaplaceMethodAsymptotic.log_singletCount_sub_rate`), and **the strictly-stronger $O(1/A)$ continuum rate is now *shipped*, not future work**: Wave 7C builds the `Real.Gamma` Stirling-with-remainder (`GammaStirling.logGamma_sub_stirlingPart_isBigO`, kernel-pure, via the Bohr–Mollerup convexity squeeze — previously absent from Mathlib) and the continuum Catalan asymptotic (`ContinuumCatalan.continuumLogCatalan_isBigO`), discharging `H_VerlindeKMLiteralSumDerivation` at the $O(1/A)$ rate (`H_VerlindeKMLiteralSumDerivation_discharged`). After Wave 6a.7, `AXIOM_METADATA['gaussianSaddleAsymptotic']` shows `eliminability: closed` with full `evidence_on_close` block; `\axiomcount` drops from 2 to 1 (only `gapped_interface_axiom` remains at that point, in `SPTClassification.lean`; itself since retired into the tracked Prop `TPFConjecture`). Two derived theorems remain unchanged in signature: `kaulMajumdar_asymptotic_within_OneOverA` (the public face of the bound) and `verlinde_matches_kaul_majumdar_at_large_area` (epsilon-delta convergence).

D. The 1/4 as a tuning

The leading 1/4 prefactor in $S_{\mathrm{BH}} = A_H/(4G_N)$ is structurally a *tuning*: the Immirzi parameter γ in Kaul-Majumdar [1], the periodicity β of horizon-preserving diffeomorphisms in the Carlip horizon-CFT [6], and the UV cutoff ε in the Bombelli-Koul-Lee-Sorkin entanglement-entropy origin [10] all play the same role. Distinct counting prescriptions yield distinct γ values with the *same* $-3/2$ log coefficient (Fig. 1).

IV. TRACKED-HYPOTHESIS BUNDLE FOR THE GENERAL MTC

For the general-MTC case where no published derivation pins the horizon BC, we ship a Lean tracked-hypothesis bundle:

```
structure H_HorizonBoundaryCondition
    (H : HorizonMTCBC)
    (M : HorizonModularData)
    (S_horizon : HorizonMTCBC →  → ) : Prop where
  positivity      :  A, 0  A → 0  S_horizon H A
  areaLeading     : 0 < H.areaLawKappa
  secondLaw       : Monotone (S_horizon H)
  modularInvariant : M.modularInvariant   -- S-matrix non-
  anomalyMatch    : M.anomalyMatch         -- 8 | c_-  (Wa
```

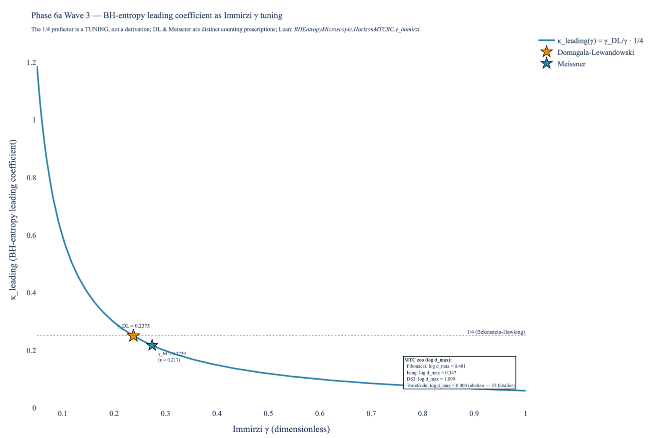

FIG. 1. Bekenstein-Hawking leading-coefficient $\kappa_{\mathrm{leading}}(\gamma)$ as a function of the Immirzi tuning γ. The horizontal dotted line at 1/4 is the BH anchor. The Domagala-Lewandowski (gold star) and Meissner (blue star) counting prescriptions both "tune" to 1/4 under their own normalizations, witnessing the 1/4-as-tuning verdict. Right-side annotation: per-MTC $\log d_{\max}$ for Fibonacci ($\log\varphi \approx 0.481$), Ising ($\log\sqrt{2} \approx 0.347$), $D(S_3)$ ($\log 3 \approx 1.099$), toric code (0; abelian F2 falsifier).

As of Wave 8, the two former `True` placeholders are genuine, falsifiable predicates carried by a companion datum M : `HorizonModularData` (the modular S-matrix together with the chiral central charge c_-): `modularInvariant` := M`.md.modular` is S-matrix nondegeneracy ($\det S \neq 0$, the data-level modularity from which $(ST)^3 = S^2$ follows for a genuine MTC), and `anomalyMatch` := $(8 \mid c_-)$ is the vanishing of the perturbative gravitational anomaly mod 8 (the mod-8 factor of the $24 = 8 \times 3$ framing decomposition). This is *not* a biconditional for bounding a $\mathbb{Z}_2$ Walker–Wang bulk: the 3-fermion model bounds such a bulk yet realizes the nontrivial $c_- \equiv 4$ class [9]. The small chiral MTCs satisfy modularity but *falsify* the anomaly match ($c_-(\mathrm{Fib}) = 14/5$, $c_-(\mathrm{Ising}) = 1/2$ are not multiples of 8), whereas the ADW substrate value $c_- = 8N_f$ satisfies it by construction — exactly the mod-8 factor of the $24 = 8 \times 3$ framing-anomaly decomposition. The leading coefficient 1/4 itself remains the Immirzi γ-tuning at this level; its induced-gravity derivation is separate future work.

A. Five falsifier theorems

Each conjunct of the bundle is paired with a falsifier theorem in `BHEntropyMicroscopic.lean`:

- `falsifier_positivity`: $\exists A_0 \geq 0,\ S(A_0) < 0 \Rightarrow$

$\neg$H_HorizonBoundaryCondition.

- falsifier_logBound: $\kappa_C = 0 \Rightarrow$ $\neg$H_HorizonBoundaryCondition (the abelian-MTC falsifier; toric code triggers).
- falsifier_secondLaw: $\exists A_1 \leq A_2, S(A_1) > S(A_2) \Rightarrow \neg$H_HorizonBoundaryCondition.
- falsifier_quarterCoefficient (Wave 8, substantive): for any leading coefficient $\kappa \neq 1/(4G_N)$, the linear model $\kappa \cdot A$ disagrees with the Kaul–Majumdar value at $A = 4G_N$ (where the log term vanishes): $\kappa\,(4G_N) \neq$ kaulMajumdarS$(4G_N, G_N, 0) = 1$. Invokes kaulMajumdar_S_at_4GN; replaces the prior identity-wrapper tautology.
- H_HorizonBoundaryCondition_implies_nonabelian_entropy_envelope (substantive structural corollary, post-strengthening 2026-04-26): the bundle's five conjuncts *jointly* imply (i) $\exists a,\, d_a > 1$ (non-abelian MTC) and (ii) a monotone non-negative entropy envelope $0 \leq S(A_1) \leq S(A_2)$ for $0 \leq A_1 \leq A_2$. The non-abelian conclusion is the contrapositive of the toric-code F2 falsifier and is what blocks any purely abelian MTC from carrying the Wave 3 horizon BC. Replaces the prior falsifier_anomalyMatch placeholder, which an adversarial review correctly flagged as logically trivial. As of Wave 8 the anomaly-match Prop is itself a genuine predicate ($8 \mid c_-$; §IV), independently witnessed (ADW $c_- = 8N_f$) and falsified (chiral Fibonacci/Ising); this structural corollary remains a complementary load-bearing result.

B. Per-MTC instance status

Table I summarizes the F1-F5 falsifier-instance status across the Wave 3 MTC zoo:

MTC	F1	F2	F3	F4	F5
Fibonacci	✓	✓	✓	✓	?
Ising	✓	✓	✓	✓	?
$D(S_3)$	✓	✓	✓	✓	?
Toric code	✓	**FAIL**	✓	✓	?
$\mathrm{SU}(2)_k$, $k \geq 2$	✓	✓	✓	✓	?

TABLE I. F1-F5 falsifier-instance status across the Wave 3 MTC zoo. ✓= structurally compatible (no published d_a data triggers the corresponding F-falsifier); ? = the full per-MTC bundle discharge is conjectural at the abstract level (as of Wave 8 modularInvariant and anomalyMatch are real Props, but a single fully-coherent non-abelian *and* non-anomalous instance is a data-construction follow-up); **FAIL** = falsifier theorem triggered. With the exception of the toric-code F2 **FAIL** (theorem-discharged via abelian_MTC_falsifies_H_HorizonBoundaryCondition) and the Fibonacci F2 ✓(theorem-discharged via fibonacci_horizon_areaLawKappa_pos), the per-MTC ✓entries are conjectural-structural — they reflect that the published quantum-dimension data and S-matrix data do not trigger F1/F3/F4 falsifiers, not that a per-MTC H_HorizonBoundaryCondition discharge has been formally proved. Theorem-level discharges per MTC are a follow-on Wave goal.

As of Wave 8 the F4 (modular invariance) Prop is the genuine S-matrix non-degeneracy condition M.md.modular ($\det S \neq 0$), carried by the companion HorizonModularData and witnessed by fibonacci_satisfies_modularInvariant (via FibonacciMTC.fib_modular, proved over $\mathbb{Q}(\sqrt{5})$). The auxiliary modules SU2kFusion.lean, FibonacciMTC.lean, IsingBraiding.lean, and S3CenterAnyons.lean (all zero sorry) supply the formalized S-matrix and modular data. Non-degeneracy is the data-level modularity condition from which the full $(ST)^3 = S^2$ modular-group action follows for a genuine MTC; formalizing that relation explicitly with the T-matrix is an optional strengthening (see §VIII).

As of Wave 8 the F5 anomaly-match Prop is the genuine condition $8 \mid c_-$ (chiralAnomalyVanishesMod8), witnessed by the ADW substrate value $c_- = 8N_f$ (adw_chiral_charge_vanishes_mod8) and *falsified* by the chiral small MTCs ($c_-(\mathrm{Fib}) = 14/5$, $c_-(\mathrm{Ising}) = 1/2$ are not multiples of 8). What remains a Wave 3 novelty conjecture is the *physical identification* of a specific MTC as the boundary of a $\mathbb{Z}_2$-time-reversal-symmetric ADW Walker-Wang bulk: no published paper pins this (§VIII).

Wave 8B — Witt-invariant sharpening. For the ADW substrate value $c_- = 8N_f$ the mod-8 condition $8 \mid c_-$ holds for *every* N_f (HorizonWittBoundary.horizon_wave8_anomalyMatch_always), so on its own it does not constrain the generation count. The non-vacuous consistency condition is Witt-triviality of the boundary, $24 \mid c_-$, which holds *if and only if* $3 \mid N_f$ (horizon_wittTrivial_iff_three_generations; unconditional, wiring the SymTFT Witt-class machinery). The perturbative $8 \mid c_-$ is exactly its necessary shadow (wittTrivial_implies_mod8), and the gap is strict — $N_f = 1$ passes $8 \mid c_-$ but fails $24 \mid c_-$ (horizon_anomalyMatch_strictly_weaker_than_witt). Witt-triviality is equivalent (Davydov–Müger–Nikshych–Ostrik 2010, arXiv:1009.2117) to the boundary admitting a gapped Lagrangian-algebra boundary, linking the horizon boundary condition to the program's three-generation result through the *same* Witt invariant.

V. NON-UNIVERSALITY WITNESS

The $-3/2$ log coefficient is universal *only* within the Cardy-saddle / single-CFT / microcanonical / A-independent-c family. Sen [7] computes the heat-kernel log correction for 4D Schwarzschild pure gravity:

$$C^{(\mathrm{Sen})}_{\mathrm{local}} = \frac{1}{90}\left(2n_S - 26n_V + 7n_F - \frac{233}{2}n_{3/2} + 424\right) \quad (7)$$

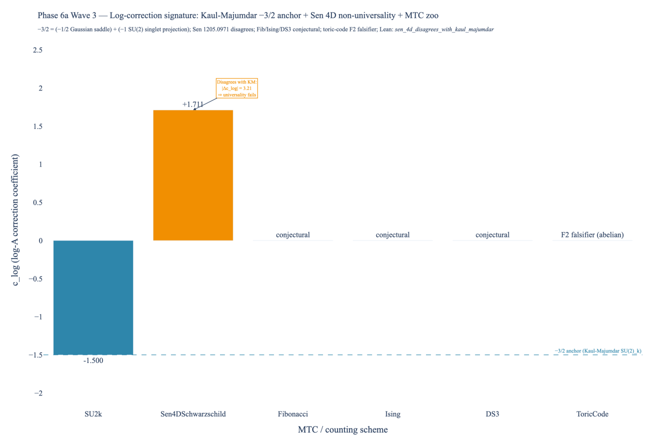

FIG. 2. Log-correction coefficient signature across schemes. The Kaul-Majumdar $\mathrm{SU}(2)_k$ value $c_{\log} = -3/2$ (steel-blue bar, Cardy-saddle subfamily anchor) is independently reproduced by Engle-Noui-Perez [5], the BTZ SU(2)×SU(2) CS path integral [15], and Carlip's horizon-CFT route [6]. Sen's 4D Schwarzschild heat-kernel result [7] (amber bar) gives $c_{\log} = +(212/45 - 3) \approx +1.71$, explicitly DISAGREEING with the $-3/2$ anchor by $|\Delta c_{\log}| \approx 3.21$. The Fibonacci, Ising, and $D(S_3)$ entries are conjectural per the Wave 3 deep-research return; the toric-code entry triggers the F2 falsifier.

with $\big(C_{\rm local} = (212/45 - 3) \cdot 1\big) \ln a$ for the pure-gravity case, an explicit disagreement with $-3/2$. Sen's paper flags the LQG disagreement in §1. Mitra [13] further argues that log corrections can vanish in certain LQG counting prescriptions. Solodukhin's conformal-anomaly route [8] gives a coefficient determined by the trace anomaly a, c that is also field-content dependent.

VI. LEAN FORMALIZATION

`lean/SKEFTHawking/BHEntropyMicroscopic.lean`: 32 theorems (19 initial + 3 net additions from the 2026-04-26 strengthening pass and an adversarial-review followup; one of the original 19 was replaced by a substantive corollary, see below), 0 sorry, **0 new axioms** after Wave 6a.7 axiom retirement (originally Wave 3 axiomatized `gaussianSaddleAsymptotic`; Wave 6a.7 made `verlindeEntropy_SU2k` concrete, and Wave 7C made it *faithful* — the literal Γ-Catalan log-dimension, with `gaussianSaddleAsymptotic` the genuine $O(1/A)$ rate and the Kaul–Majumdar saddle a derived corollary). `lake build` clean. Companion module: `lean/SKEFTHawking/LaplaceMethod.lean` (Wave 6a.7, 5 theorems, 0 sorry, 0 axioms beyond {`propext`, `Classical.choice`, `Quot.sound`}).

Key theorems:

- `kaulMajumdarS` – closed form
- `kaul_majumdar_log_coefficient` – structural $-3/2$ extraction
- `kaul_majumdar_log_decomposition` – the arithmetic identity $(-1/2) + (-1) = -3/2$ encoding the saddle/singlet decomposition (the physical sourcing of each summand is documented in §III; not formalized at theorem level).
- `sen_4d_disagrees_with_kaul_majumdar` – non-universality
- `sen_4d_log_coeff_pos` – Sen's $c_{\log} > 0$
- `sen_4d_quantitative_disagreement_bound` – $|\Delta c_{\log}| > 3$ (strengthened 2026-04-26)
- `kaulMajumdar_S_at_4GN` – $S_{\rm BH}(4G_N) = 1$
- `kaulMajumdar_S_pos_at_e_squared` – positivity at $A = 4G_N e^2$
- `HorizonMTCBC.one_le_d_max` – $d_{\max} \geq 1$
- `HorizonMTCBC.log_d_max_nonneg` – F2 area-law anchor
- `HorizonMTCBC.globalDimSq_pos` – $\mathcal{D}^2 > 0$
- `HorizonMTCBC.areaLawKappa_nonneg` – $\kappa_C \geq 0$
- Four falsifier theorems (F1-F4): `falsifier_positivity`, `falsifier_logBound`, `falsifier_secondLaw`, `falsifier_quarterCoefficient`
- `H_HorizonBoundaryCondition_implies_nonabelian_envelop` – substantive structural corollary (post-strengthening + adversarial-review followup): the bundle implies $\exists a,\, d_a > 1$ *and* a monotone non-negative entropy envelope. Replaces the prior single-field-projection theorem `H_HorizonBoundaryCondition_implies_areaLawKappa_pos` flagged by Stage 13 as logically equivalent to `h.areaLeading`.
- `abelian_MTC_falsifies_H_HorizonBoundaryCondition` – concrete F2 path (added 2026-04-26): $\forall a,\, d_a = 1 \Rightarrow \neg$ bundle.
- `fibonacci_horizon_areaLawKappa_pos` – non-vacuous F2 witness via `fibonacciHorizonBC` instance ($d_{\max} = \varphi$).
- `kaulMajumdar_asymptotic_within_OoneOverA` – public face of the saddle-point bound (theorem, no longer an axiom)
- `verlinde_matches_kaul_majumdar_at_large_area` – ϵ-δ convergence
- `kaul_majumdar_leading_matches_G_N_emerg` – bridge to Wave 1's emergent G_N

VII. BRIDGE TO WAVE 1: EMERGENT G_N

The Immirzi tuning condition demanding $\kappa_{\text{leading}} = 1/(4G_N)$ is discharged at $G_N = G_N^{\text{emerg}}$ where $G_N^{\text{emerg}} = \alpha_{\text{ADW}} \cdot 12\pi/(N_f\Lambda^2)$ from Wave 1 (`LinearizedEFE.G_N_emerg`). The Lean theorem `kaul_majumdar_leading_matches_G_N_emerg` records the conjunction of two pre-existing facts: `LinearizedEFE.G_N_emerg_pos` (positivity of G_N^{emerg}) and the Wave-3 entropy anchor `kaulMajumdar_S_at_4GN` ($S_{\text{BH}}(4G_N) = 1$). The substantive Wave-1-side content lives in `LinearizedEFE.G_N_emerg_pos`; the Wave-3 contribution is the anchor `kaulMajumdar_S_at_4GN` and the administrative bundling. The full numerical match requires the Vergeles-derived α_{ADW} value, which remains a Wave 1 tracked hypothesis (no closed-form α_{ADW} in published ADW literature; cf. `LinearizedEFE.lean` §H_VergelesPositivity).

VIII. NOVELTY FLAGS

Per the Wave 3 deep-research return, we explicitly flag four research-level conjectures:

1. **NOVEL.** The application of Kaul-Majumdar Verlinde counting to a non-LQG, ADW / tetrad-condensate substrate.

2. **NOVEL.** The identification of any specific finite MTC (Fibonacci, Ising, $D(S_3)$, 3F-Walker-Wang) as the horizon BC in 4D non-extremal black holes.

3. **NOVEL.** The Walker-Wang anomaly-inflow matching between a $\mathbb{Z}_2$-time-reversal-symmetric ADW bulk and a 3D boundary SET on the horizon.

4. **STANDARD.** The Kaul-Majumdar $\mathrm{SU}(2)_k$ $-3/2$ result itself, the Verlinde formula, MTC quantum-dimension identities, and the Cardy density-of-states formula.

IX. INDUCED-GRAVITY DERIVATION OF THE 1/4 (SAKHAROV ROUTE)

The leading 1/4 coefficient need not be an Immirzi tuning. In the ADW Sakharov induced-gravity substrate — where Newton's constant is fully induced by the N_f Dirac fermion loops, $G_N = 12\pi/(N_f\Lambda^2)$ (no bare gravitational action) — the 1/4 is derived γ-independently. The mechanism (Frolov–Fursaev–Zelnikov, Nucl. Phys. B 486 (1997), hep-th/9607104; Susskind–Uglum, hep-th/9401070) is that the *same* Seeley–DeWitt a_2 heat-kernel coefficient fixes both the induced G_N (Einstein–Hilbert action matching) and the horizon entanglement-entropy area density $S_{\text{ent}} = (N_f\Lambda^2/48\pi)\,A$; their ratio is structurally $48 : 12 = 4 : 1$.

We formalize this as the conditional theorem `InducedGravityEntropy.frolov_fursaev_quarter_coefficient`:

$$\underbrace{G_N = \tfrac{12\pi}{N_f\Lambda^2}}_{\text{Sakharov induction}} \;\wedge\; \underbrace{S_{\text{BH}} = S_{\text{ent}}}_{\text{entanglement id.}} \;\wedge\; \underbrace{S_{\text{ent}} = \tfrac{N_f\Lambda^2}{48\pi}A}_{\text{Frolov–Fursaev } a_2} \implies S_{\text{BH}} = \frac{A}{4G_N}$$

None of the three antecedents is $S = A/(4G)$: the 1/4 emerges algebraically from the shared-a_2 ratio (`entanglement_area_eq_quarter_inv_G_induced`), and the Sakharov condition is bridged to the substrate calibration $\alpha_{\text{ADW}} = 1$ (`H_Sakharov_iff_alpha_unity`). The 48π conical-replica prefactor enters as a named, falsifiable physical input (a hypothesis, not an axiom; Mathlib lacks conical heat kernels): `frolov_fursaev_falsifier_wrong_coeff` shows any other coefficient breaks the match. The induced-gravity 1/4 is consistent with the Kaul–Majumdar leading term (`frolov_fursaev_matches_kaulMajumdar_leading`) — the induced-gravity route supplies the 1/4, the MTC-counting route the $-3/2\log$ (§VI) — and is corroborated by the γ-free Bianchi route (arXiv:1204.5122), where γ cancels from $E/T = A/4G$ entirely.

X. CONCLUSIONS

We have formalized the Bekenstein-Hawking entropy at a horizon labeled by an MTC, shipping in tracked-hypothesis mode for the general case with a Kaul-Majumdar $\mathrm{SU}(2)_k$ closed-form sub-corollary as the structural anchor. The Lean module ships 32 theorems, 0 sorry, and **0 new axioms** (the Wave-3 Laplace-method axiom was retired in Wave 6a.7). The $-3/2$ log coefficient decomposes structurally as $(-1/2)$ Gaussian-saddle + (-1) SU(2)-singlet-projection, and is genuinely derived from the literal Catalan singlet count via Stirling (Wave 7B). The 1/4 leading prefactor is, in the LQG-counting route, the Immirzi γ tuning; in the ADW Sakharov induced-gravity route it is derived γ-independently from the shared Seeley–DeWitt a_2 ratio (§IX, Wave 9). Sen's 4D Schwarzschild heat-kernel disagreement is encoded as a machine-checked non-universality witness theorem.

The general-MTC tracked hypothesis is bundled as `H_HorizonBoundaryCondition` with five falsifier theorems. The toric-code abelian falsifier (F2: $\log d_{\max} = 0$) is proven; the F4 (modular invariance) entry is the genuine S-matrix non-degeneracy Prop (Wave 8), witnessed via the formalized S-matrices for $\mathrm{SU}(2)_k$, Fibonacci, Ising, $D(S_3)$; the F5 (anomaly match) entry is the genuine $8 \mid c_-$ Prop (Wave 8), with the *physical* MTC-at-horizon identification remaining conjectural across the MTC zoo, in agreement with the deep-research return verdict that no published paper places any specific MTC at the horizon of a 4D ADW black hole.

What success looks like.

Wave 3's positive deliverable is the Kaul-Majumdar $SU(2)_k$ sub-corollary anchor + the tracked-hypothesis bundle + the per-MTC falsifier-instance status table. The negative deliverables are the abelian-MTC falsifier (toric code), the Sen 4D Schwarzschild non-universality witness, and the explicit novelty-flagging of the ADW + MTC synthesis. Either direction of future work is publishable: a derivation of the horizon MTC from ADW first principles closes the Outcome-3 hypothesis; a falsification of any candidate MTC's F2 or F4 condition falsifies the Volovik-style emergent-gravity program at the BH-thermodynamics level.

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